Shajid Hossain Hemal

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Experience

Samsung R&D Institute Bangladesh (SRBD) Dhaka, Bangladesh
Lead Engineer March 2026 – Present

- Leading the development of a Physical AI Simulator framework for multiple autonomous humans in NVIDIA Isaac Sim, integrating navigation, animation, and behavior systems.
- Designed indoor simulation environments supporting walking, sitting, and pick-and-place interactions, enabling controllable scenarios for Physical AI and embodied AI research.
- Implemented a crowd avoidance system to prevent inter-agent collisions, enabling realistic large-scale multi-human simulations with coherent spatial awareness within the Physical AI environment.
- Driving technical direction and architecture decisions for the team, ensuring alignment with long-term product goals.

Senior Software Engineer March 2024 – February 2026

- Designed and implemented the AI Cover Feature to auto-generate visual note summaries from user notes.
- Led frontend development for the collaborative editing feature, enabling real-time multi-user interactions.
- Re-architected the application using WinUI 3 with C++/WinRT, significantly improving maintainability and future readiness.
- Optimized the main list view's loading time and scrolling performance, achieving 15% improvement in startup time and 30% enhancement in scrolling — recognized with the Icon award.
- Contributed to multiple AI-driven features in collaboration with Gemini 1.5 Flash, enhancing functionality and automation within the app.
- Mentored junior developers through code reviews, debugging sessions, and design decisions, raising overall team code quality.

Software Engineer June 2022 – February 2024

- Engineered and delivered robust features in list, search, sort, and filter views using modern UI/UX practices.
- Improved app efficiency by implementing optimized thumbnail generation and robust file/data management logic.
- Collaborated on the design and development of the multi-instance feature, enhancing multitasking capability.
- Diagnosed and fixed 300+ bugs, contributing to major version releases and stability improvements.

Skills

Languages: C#, C++, WinRT/C++, Python
Frameworks & UI: WinUI3, UWP, WPF, MVVM, MVPVM, SOLID
AI & ML: NVIDIA Isaac Sim, PyTorch, OpenCV, NumPy, SQLite
Tools & DevOps: Git & GitHub, Perforce, Unity

Education

Noakhali Science and Technology University Noakhali, Bangladesh
Bachelor of Science in Information and Communication Engineering January 2017 – April 2022
CGPA – 3.68 out of 4.00

Projects

Physical AI Simulator *Python, NVIDIA Isaac Sim, PyTorch, OpenCV, NumPy*
A simulation framework for multiple autonomous humans in NVIDIA Isaac Sim, integrating navigation, animation, and behavior systems. Designed indoor environments supporting walking, sitting, and pick-and-place interactions, providing controllable scenarios for Physical AI and embodied AI research.

Windows Samsung Notes *UWP, WPF, WinUI3, C#, WinRT/C++, MVVM, MVPVM, SOLID* | Store
A feature-rich note-taking app by Samsung Electronics with millions of users on Windows. Owned and delivered the main list view with optimized loading and scrolling performance (15% faster startup, 30% smoother scrolling), the AI Cover feature that auto-generates visual note summaries, the thumbnail generation pipeline, and the Tape Pen inking feature.

Awards & Achievements

Icon Award *July/August 2025* | Award
Led WinUI 3 migration and delivered the AI Cover feature, advancing the app's architecture and capabilities.

Icon Award *March 2024* | Award
Achieved 15% improvement in app startup time and 30% enhancement in note list scrolling performance.

Samsung Software Competency Professional Certificate *January 2024* | Certificate
Passed Samsung's globally administered Professional-level assessment in algorithms and data structures.

Publications

Predicting the Impact of Internet Usage on Students' Academic Performance Using Machine Learning Techniques in Bangladesh Perspective *Mar 2024*
Analyzes the effects of internet usage on students' academic progress and predicts performance using machine learning algorithms. Offers practical applications and policy recommendations for Bangladesh's education sector. [Paper Link]